

ASSIGNMENT 7

Title: Write an Embedded C program for generating PWM signal for servo motor/DC motor.

Objective: To learn interfacing of servo motor with PIC18FXXXX microcontroller

Problem statement: Write an Embedded C program for generating PWM signal for servo motor/DC motor.

Input:

```
#include<p18f4550.h>
```

```
unsigned char count=0;
```

```
bit TIMER,SPEED_UP;
```

```
void timer2Init(void)
```

```
{
```

```
    T2CON    =    0b00000010;           //Prescalar = 16; Timer2 OFF
```

```
    PR2      =    0x95;                 //Period Register
```

```
}
```

```
void delay(unsigned int time)
```

```
{
```

```
    unsigned int i,j;
```

```
    for(i=0;i<time;i++)
```

```
        for(j=0;j<1000;j++);
```

```
}
```

```

void main(void)
{
    unsigned int i;
    TRISCbits.TRISC1    = 0;           //RC1 pin as output
    TRISCbits.TRISC2    = 0;           //CCP1 pin as output
    LATCbits.LATC1      = 0;
    CCP1CON = 0b00111100;             //Select PWM mode; Duty cycle LSB
    CCP1CON<4:5> = <1:1>
    CCPR1L = 0x0F;                     //Duty cycle 10%
    timer2Init();                     //Initialise Timer2
    TMR2ON = 1;                        //Timer2 ON

    while(1)                           //Loop forever
    {
        for(i=15;i<150;i++)
        {
            CCPR1L = i;
            delay(100);
        }
        for(i=150;i>15;i--)
        {
            CCPR1L = i;
            delay(100);
        }
    }
}

```